

Control Systems Project Summary

Key Objectives

- Analyze and model an electrical system in the time domain using differential equations.
- Develop and simulate the system in Simulink to observe transient behavior.
- Design and apply a PI controller to improve system performance.
- Compare open-loop and closed-loop system responses using specific metrics.
- Evaluate the effectiveness of compensation based on steady-state error, overshoot, and settling time.

Technical Overview

This project was conducted as part of Control Systems 3 and involved the mathematical modeling and simulation of an RLC circuit. The transfer function was derived analytically using Laplace transforms. Simulation of the open-loop system showed poor transient performance and high steady-state error. To improve this, a PI controller was designed using proportional (K_p) and integral (K_i) gains. The compensated system was analyzed and showed improved settling time, reduced error, and acceptable overshoot. Comparisons were made between the baseline and controlled system to demonstrate improvements in dynamic response.

Visual Aids

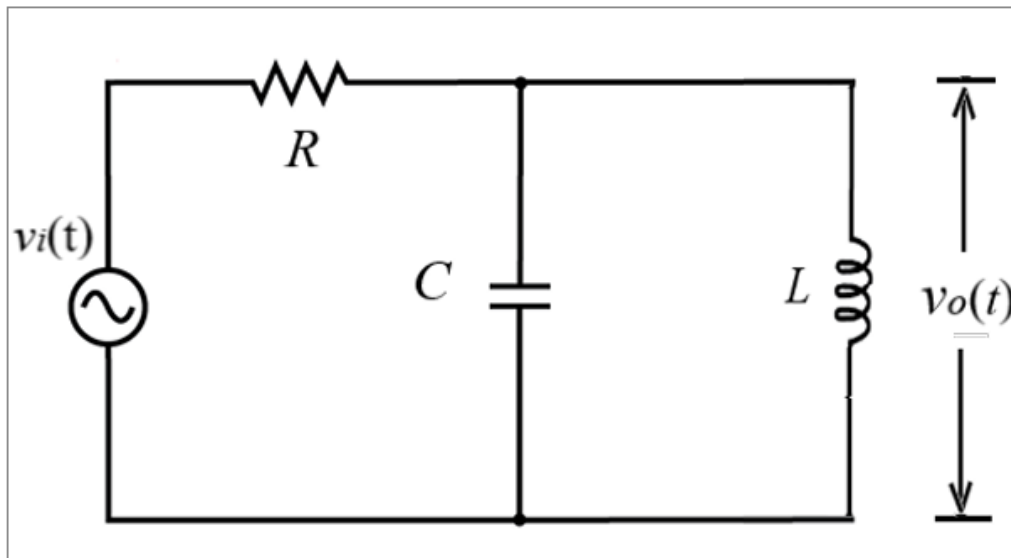


Figure 1: A basic electrical system model for system response analysis.

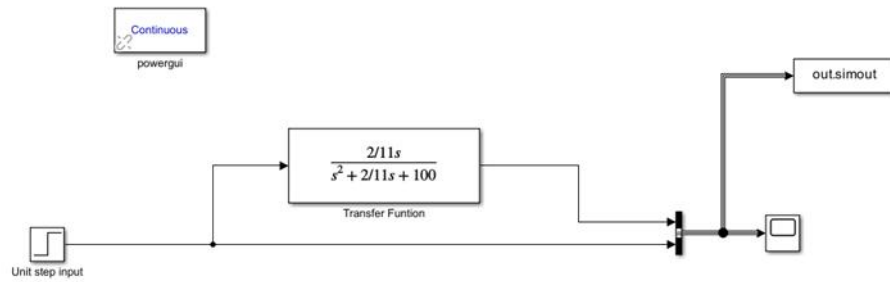


Figure 2: Simulation of PI controller derived from Laplace Transform for Section A

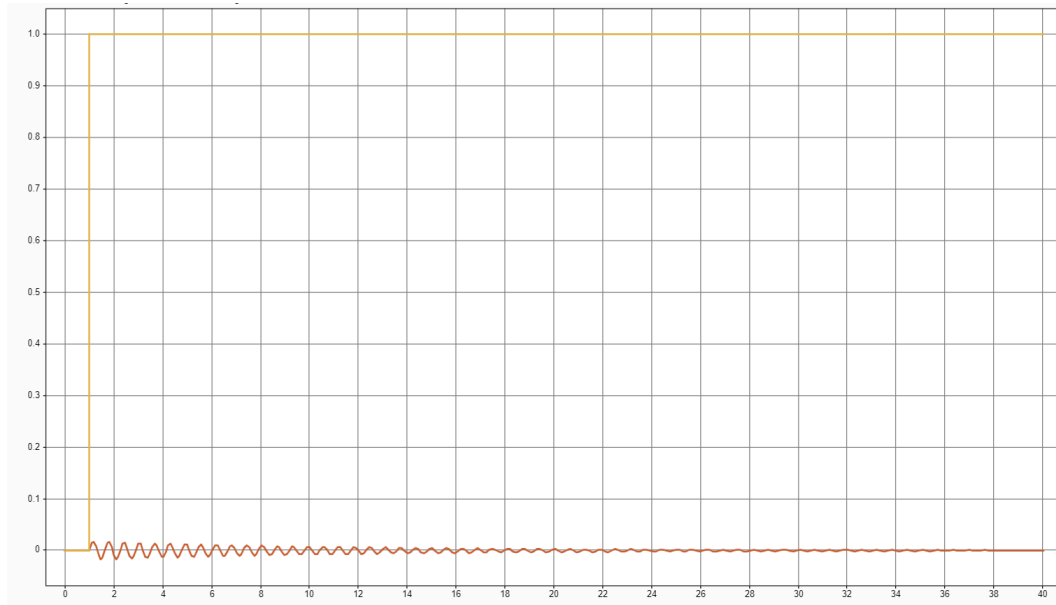


Figure 3: Poor input and output response from oscilloscope for Section A.

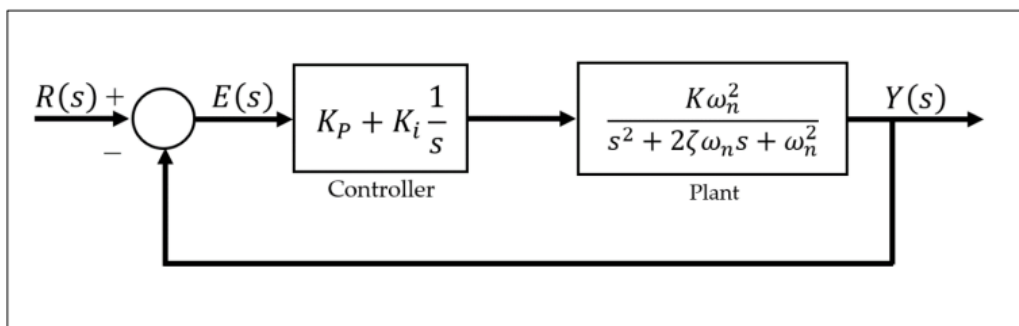


Figure 4: A PI controller design for compensation of the system response

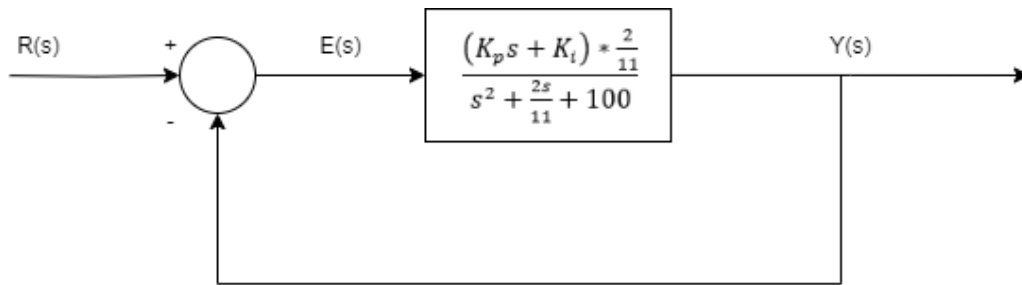


Figure 5: closed loop system for the system model

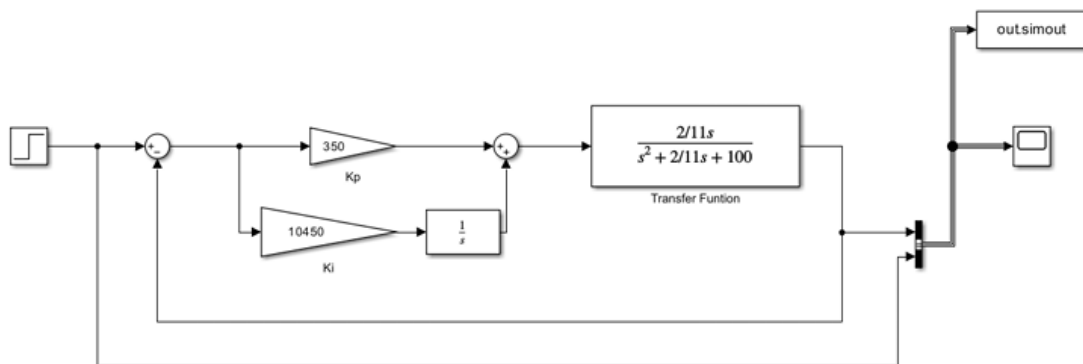


Figure 6: Design of a system model in MATLAB Simulink for system with controller gain – Section B

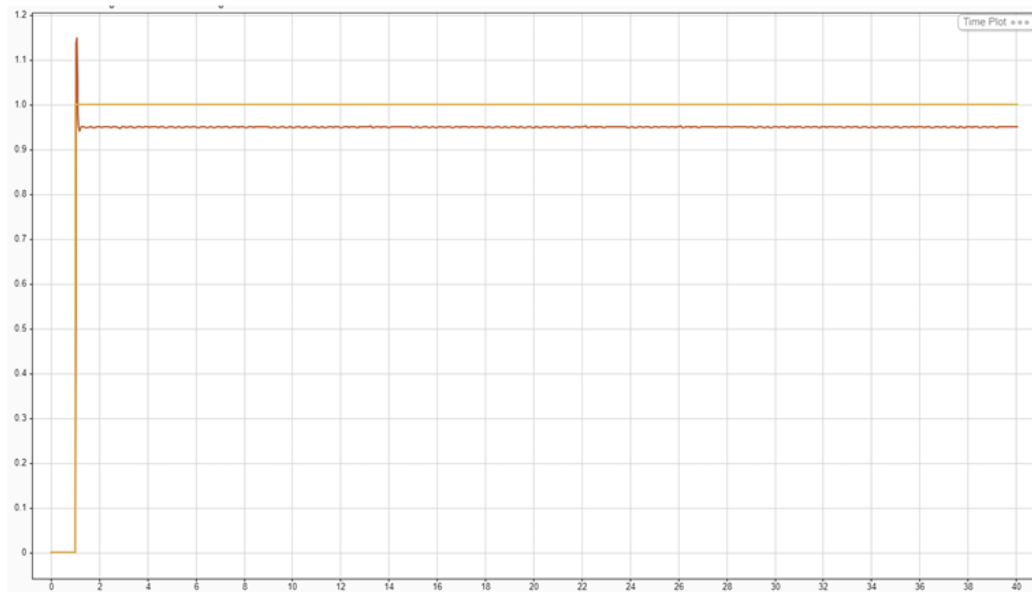


Figure 7: Improved input and output response from oscilloscope for Section B

Parameter	Section A result	Section B result
Transient response	Underdamped system with many oscillations	Underdamped system with less oscillations
Rise time	$t_r = 0.1580609734s$	$t_r = 0.08069000515s$
Settling time	$t_s = 33s$	$t_s = 0.09401705402s$
Steady-state error	$e_{ss} = 100\%$	$e_{ss} = 5\%$
System overshoot	PO (%) = 0%	PO (%) = 17.1%

Table 1: Comparison of parameter results between Section A and Section B systems

Highlights

- Successfully modeled an RLC circuit and derived the transfer function.
- Simulated system behavior using Simulink and evaluated time domain metrics.
- Designed a PI controller to meet specified criteria ($\leq 20\%$ overshoot, $\leq 5\%$ error).
- Achieved a significant improvement in system stability and response time.
- Comparative analysis clearly demonstrated the benefits of closed-loop control.

Conclusion

The project effectively demonstrated the use of engineering tools to model, simulate, and improve control system performance. The PI controller met all performance requirements, reducing steady-state error to 5% and limiting overshoot to 17.1%, while drastically improving the system's settling time. This validates the value of control strategies in practical system design.